

BMKG STRONG MOTION ANALYZER

Laporan Analisis Strong Motion

1. RINGKASAN GUNCANGAN TERKUAT

Stasiun TKJN _ 2026-02-05 18_04_38 UTC mencatat nilai percepatan tanah maksimum (PGA) sebesar 0.374163 m/s² atau 37.416 Gal pada komponen HNN. Berdasarkan klasifikasi SIG-BMKG yang digunakan oleh BSMA, hasil tersebut termasuk Skala II (DIRASAKAN) dengan rentang intensitas III-V.

Network	IA
Station	TKJN
Location	
Sampling Rate	100.000 Hz
Start Time	2026-02-05T18:04:38.510000Z
End Time	2026-02-05T18:10:14.950000Z
Channels	HNE, HNN, HNZ

SKALA II SIG-BMKG | DIRASAKAN

Warna: Hijau | Rentang intensitas: III-V

Dampak: Dirasakan oleh orang banyak tetapi tidak menimbulkan kerusakan. Benda-benda ringan yang digantung bergoyang dan jendela kaca bergetar.

INSTRUMENTAL INTENSITY (MMI): V | PERCEIVED SHAKING: MODERATE

Potensi Kerusakan: Very light | Peak Acc Baseline: 6.2 %g | Peak Vel Baseline: 4.7 cm/s

Nilai Rekaman Terukur: PGA = 3.815 %g | PGV = 2.119 cm/s (Skala USGS ShakeMap / Worden et al., 2011)

2. PARAMETER STRONG MOTION

Channel	PGA (Gal)	PGV (cm/s)	PGD (cm)	Arias (m/s)	D5-95 (s)	SIG
HNE	36.849	2.7863	0.3114	0.008642	23.72	II (III-V)
HNN	37.416	2.1187	0.2904	0.009056	21.65	II (III-V)
HNZ	8.627	0.9081	0.1786	0.002183	48.25	II (III-V)

Parameter komponen terkuat:

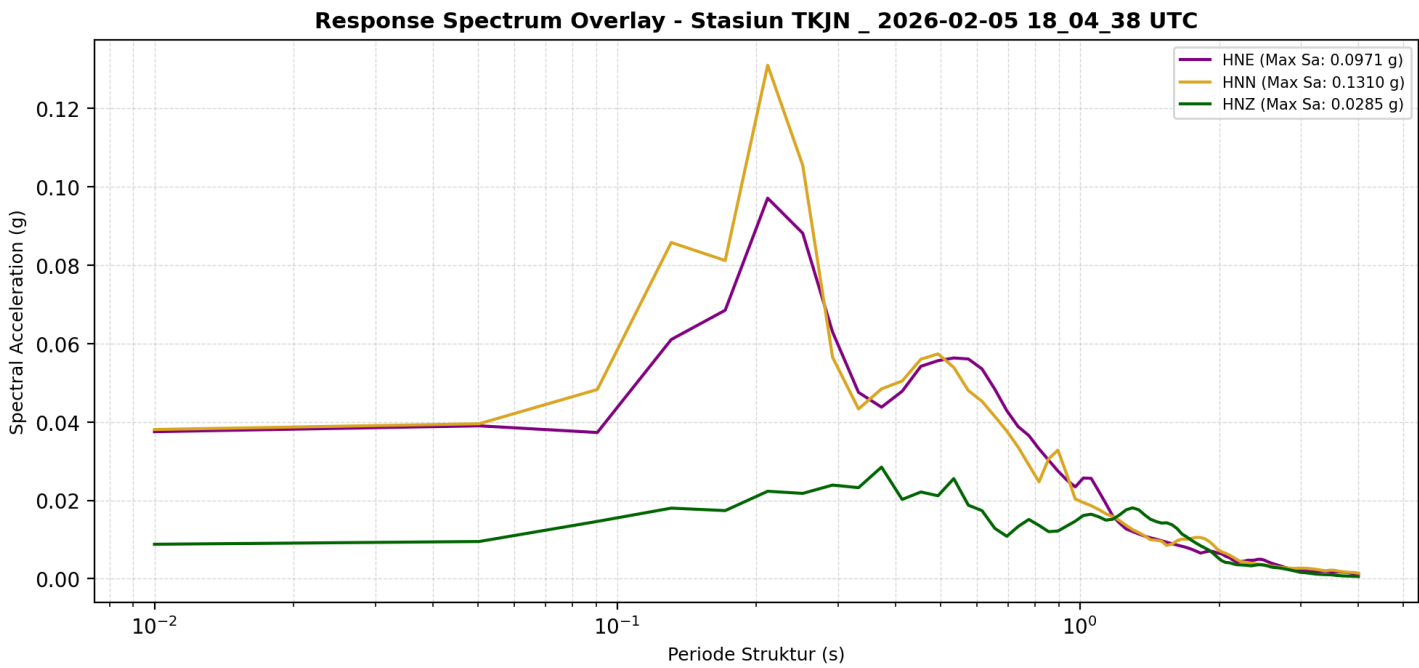
PGA = 0.374163 m/s² (37.416 Gal); PGV = 2.1187 cm/s; PGD = 0.2904 cm; Arias Intensity = 0.009056 m/s; D5-95 = 21.65 s.

BMKG STRONG MOTION ANALYZER

Laporan Analisis Strong Motion

3. RESPONSE SPECTRUM

Kurva response spectrum menunjukkan percepatan spektral sebagai fungsi periode struktur berdasarkan hasil pemrosesan response spectrum pada pipeline BSMA.

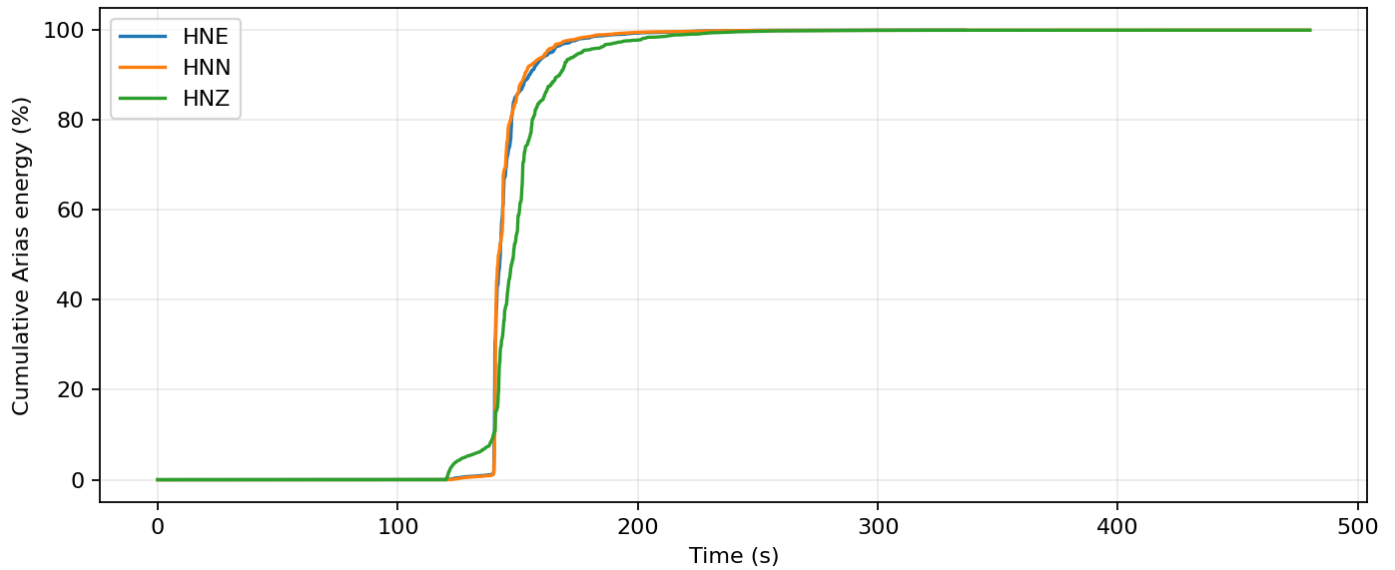


BMKG STRONG MOTION ANALYZER

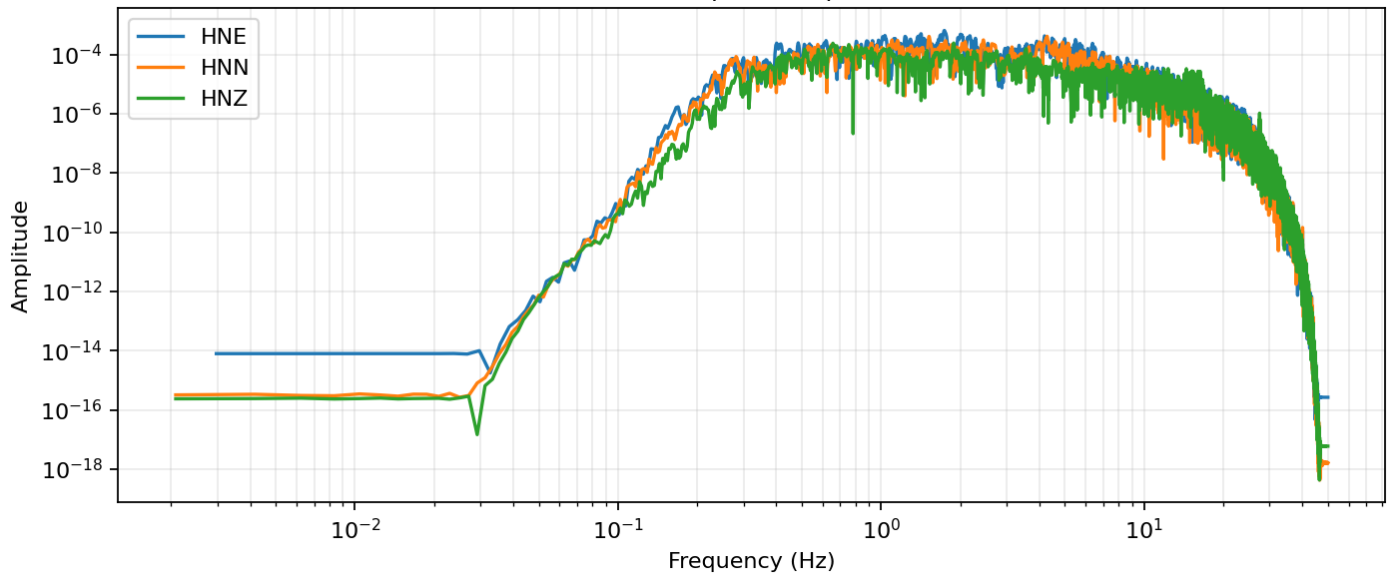
Laporan Analisis Strong Motion

4. HUSID PLOT DAN FOURIER AMPLITUDE SPECTRUM

Husid energy curve



Fourier amplitude spectrum



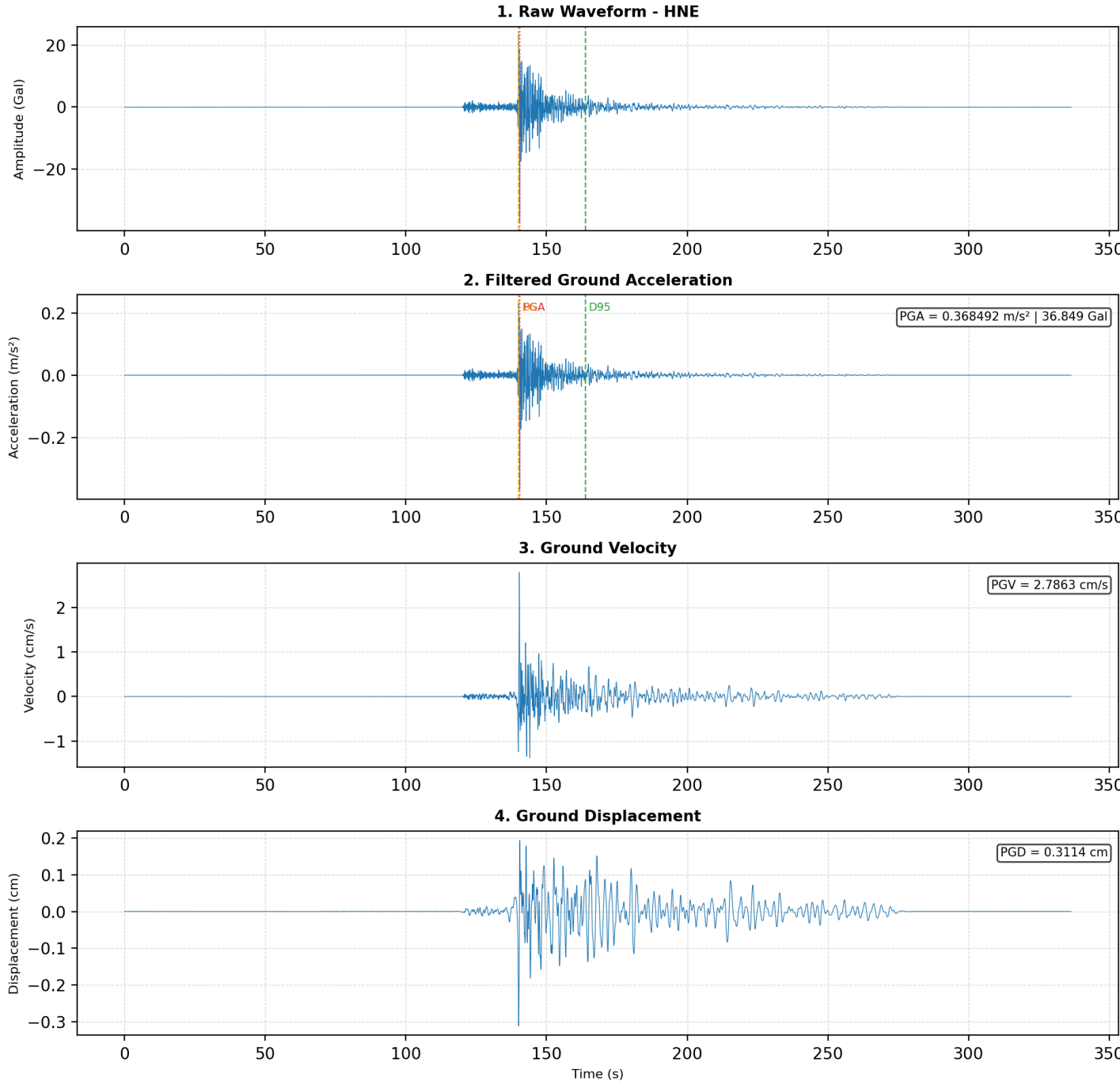
BMKG STRONG MOTION ANALYZER

Laporan Analisis Strong Motion

5. WAVEFORM ANALYSIS - HNE

PGA: 36.849 Gal | PGV: 2.7863 cm/s | PGD: 0.3114 cm | D5-95: 23.72 s

Analisis Kinematik Ground Motion
Stasiun: TKJN _ 2026-02-05 18_04_38 UTC | Channel: HNE



BMKG STRONG MOTION ANALYZER

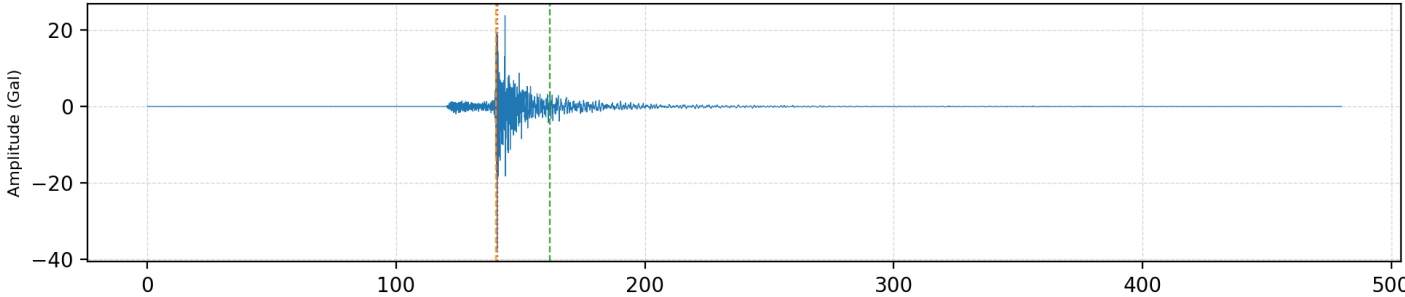
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5. WAVEFORM ANALYSIS - HNN

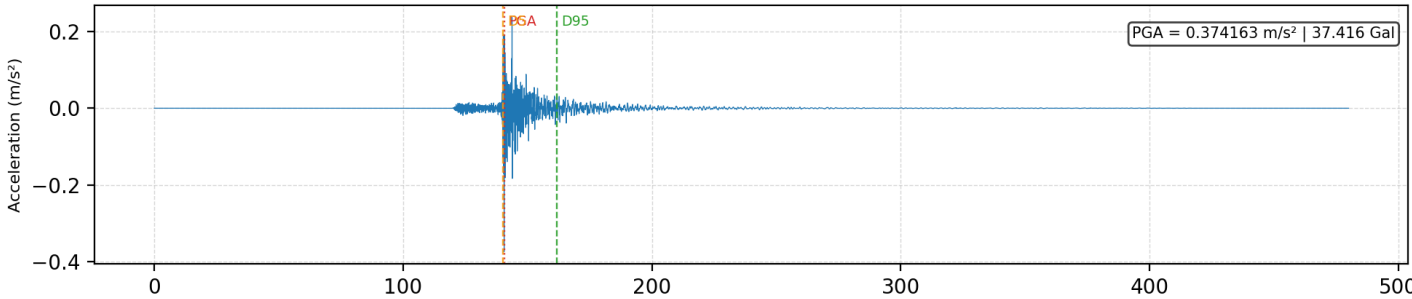
PGA: 37.416 Gal | PGV: 2.1187 cm/s | PGD: 0.2904 cm | D5-95: 21.65 s

Analisis Kinematik Ground Motion
Stasiun: TKJN _ 2026-02-05 18_04_38 UTC | Channel: HNN

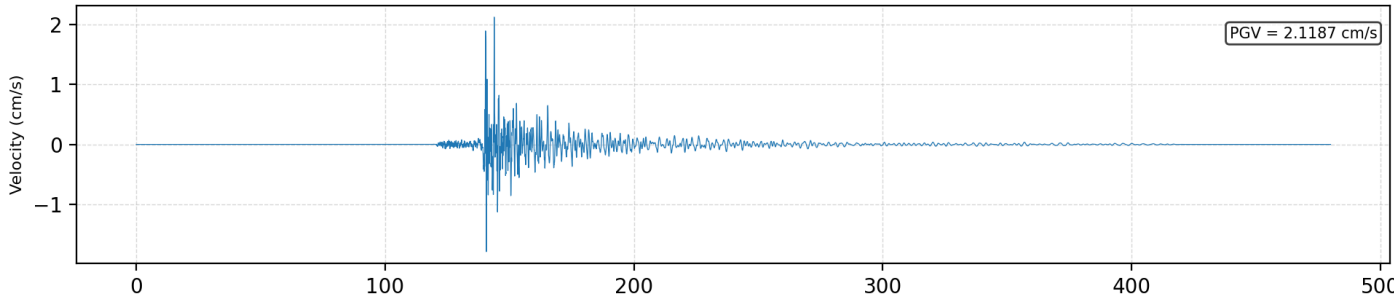
1. Raw Waveform - HNN



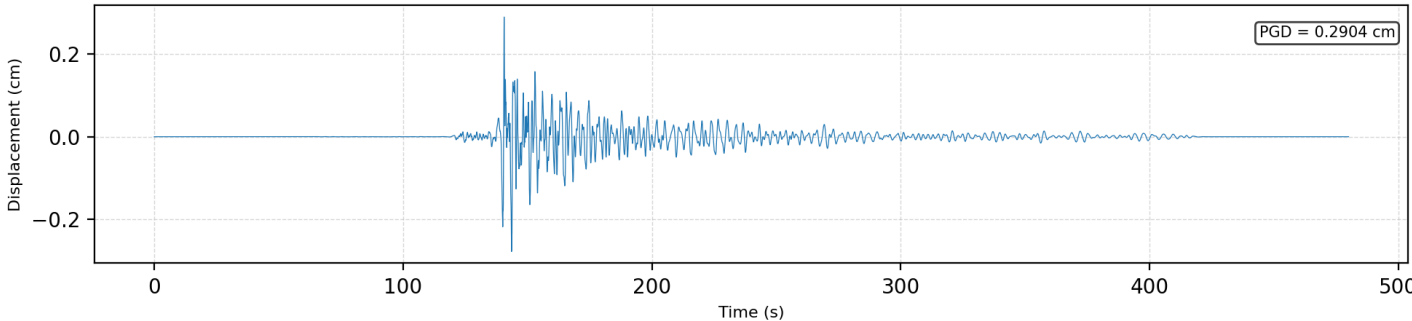
2. Filtered Ground Acceleration



3. Ground Velocity



4. Ground Displacement



BMKG STRONG MOTION ANALYZER

Laporan Analisis Strong Motion

5. WAVEFORM ANALYSIS - HNZ

PGA: 8.627 Gal | PGV: 0.9081 cm/s | PGD: 0.1786 cm | D5-95: 48.25 s

Analisis Kinematik Ground Motion
Stasiun: TKJN _ 2026-02-05 18_04_38 UTC | Channel: HNZ

